Problem 1

Given the following,

$$E(W, v|X) = -\sum_{t=1}^{N}[r^t log(y^t) + (1 - r^t)log(1 - y^t)] + \sum_{h=1}^{H} ||w_h||_2^2$$

$$y^t = sigmoid(\sum_{t=1}^{H} v_h z_h^t + v_0)$$

$$z_h^t = reLU(w_h^T x^t + w_{h0})$$

We will say that

$$\alpha = w_h^T x^t + w_{h0}$$

and

$$\beta = \sum_{t=1}^{H} v_h z_h^t + v_0$$

and derive the following update equations:

$$\frac{\delta E}{\delta w_{hj}} = \frac{\delta E}{\delta y^t}\frac{\delta y^t}{\delta \beta}\frac{\delta \beta}{\delta z_h^t}\frac{\delta z_h^t}{\delta \alpha}\frac{\delta \alpha}{\delta w_{hj}}$$

$$\frac{\delta E}{\delta w_{h0}} = \frac{\delta E}{\delta y^t}\frac{\delta y^t}{\delta \beta}\frac{\delta \beta}{\delta z_h^t}\frac{\delta z_h^t}{\delta \alpha}\frac{\delta \alpha}{\delta w_{h0}}$$

$$\frac{\delta E}{\delta v_h} = \frac{\delta E}{\delta y^t}\frac{\delta y^t}{\delta \beta}\frac{\delta \beta}{\delta v_h}$$

$$\frac{\delta E}{\delta v_0} = \frac{\delta E}{\delta y^t}\frac{\delta y^t}{\delta \beta}\frac{\delta \beta}{\delta v_0}$$

We first proceed to find the gradient for the hidden layer's weights.

$$\frac{\delta E}{\delta w_{hj}} = (-\sum_{t=1}^{N}\frac{r^t}{y^t} - \frac{1 - r^t}{1 - y^t})(y(1 - y))(\sum_{h=1}^{H} v_h)(x^t) + 2w$$

$$\frac{\delta E}{\delta w_{hj}} = (-\sum_{t=1}^{N}\frac{r^t - r^t y^t - y^t + r^t y^t}{y^t - y^{2t}})(y(1 - y))(\sum_{h=1}^{H} v_h)(x^t) + 2w$$

$$\frac{\delta E}{\delta w_{hj}} = (-\frac{r-y}{y-y^2})(y-y^2)(v)(x^t) + 2w$$

$$\frac{\delta E}{\delta w_{hj}} = (r-y)(v)(x) + 2w$$

Then, we use a similar process to find the gradient for the hidden layer's bias. Since this is similar to the weights, we only need to change the final partial derivative to differentiate α with respect to w_0 instead of w.

$$\frac{\delta E}{\delta w_{h0}} = (r-y)(v)$$

Now, we proceed to derive the gradients for $\frac{\delta E}{\delta v_h}$ and $\frac{\delta E}{\delta v_0}$

$$\frac{\delta E}{\delta v_h} = (-\sum_{t=1}^{N} \frac{r^t}{y^t} - \frac{1-r^t}{1-y^t})(y(1-y))(\sum z_h^t)$$

$$\frac{\delta E}{\delta v_h} = (-\frac{r-y}{y-y^2})(y-y^2)(z)$$

$$\frac{\delta E}{\delta v_h} = (-\frac{r-y}{y-y^2})(y-y^2)(z)$$

$$\frac{\delta E}{\delta v_h} = -(r-y)(z)$$

Once again, the bias for the output layer is also related to the weights except for the last partial derivative which created the z term in the output layer weights, so the bias is similar:

$$\frac{\delta E}{\delta v_0} = (-\sum_{t=1}^{N} \frac{r^t}{y^t} - \frac{1-r^t}{1-y^t})(y(1-y))$$

$$\frac{\delta E}{\delta v_0} = -(r-y)$$

Problem 2

For figure 2, $w = [1, 1]^T$

For figure 3, $w = [-1, 1, 1]^T$

For figure 5, $z_1 = S(x_2 + 1)$ and $z_2 = S(x_1 + x_2 - 1)$ so $y = S(z_1 + z_2 - 1.5)$ is the final activation function. $\mathbf{W} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix}^T$ and $v = \begin{bmatrix} -1.5 & 1 & 1 \end{bmatrix}$

Problem 3

Report the validation accuracy by the number of hidden units. How many hidden units should we use? Report the accuracy on the test set using this number of hidden units.

Validation accuracy for 4 hidden units is 0.707
Validation accuracy for 16 hidden units is 0.884
Validation accuracy for 20 hidden units is 0.912
Validation accuracy for 24 hidden units is 0.914
Validation accuracy for 32 hidden units is 0.917
Validation accuracy for 48 hidden units is 0.923
Test accuracy with 48 hidden units is 0.102

The number of hidden units ended up being 48. This however resulted in a test accuracy of only 0.102 probably as a result of overfitting to the model.

Compare the 2-D and 3-D plots and explain the results in the report.

The hidden layers allowed us to consider this problem in a non-linear way. Similarly to how using a two layer perceptron allowed us to quantify XOR, we need a non-linear MLP to help classify the 10 different digits. The difference between 2D and 3D then is that 3D should result in better separation of classes.

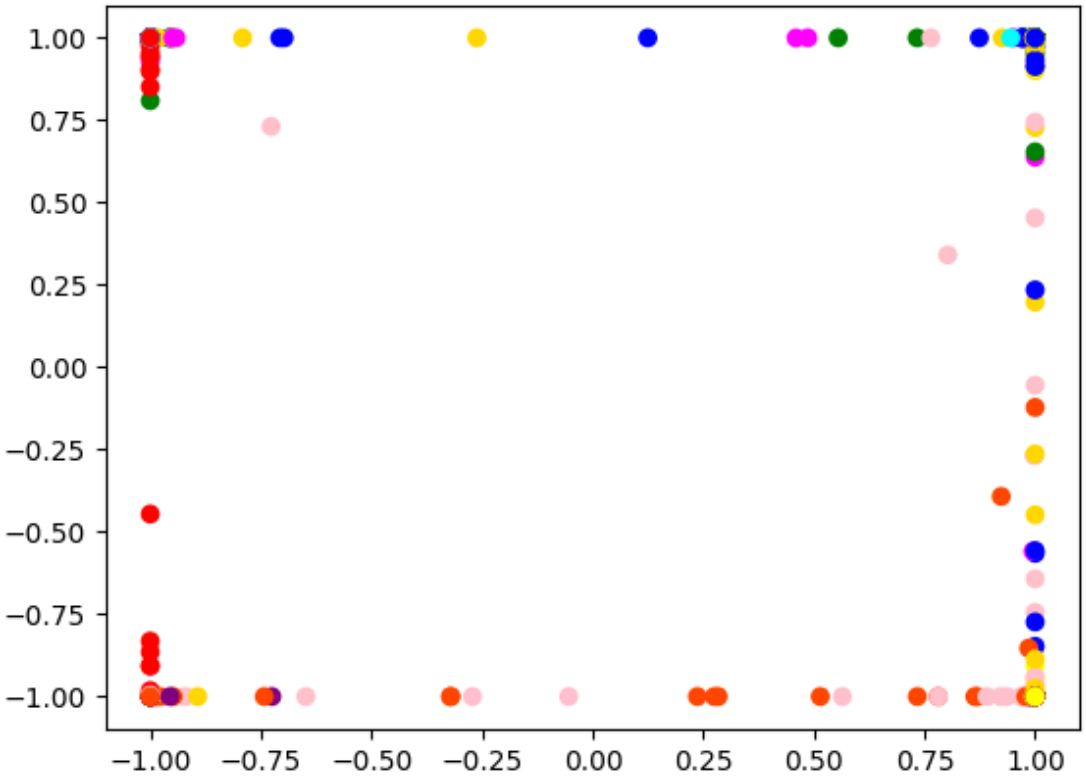

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

Figure 1: 2D

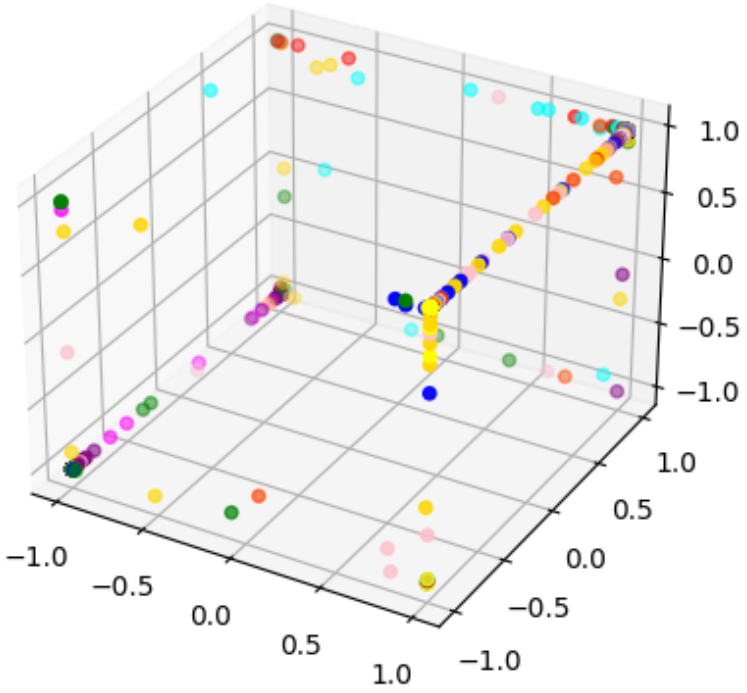


Figure 2: 3D